6.1 Numbers to 100,000: activities

← Unit overview

1. Big numbers around us

You need: a newspaper, a supermarket folder, a phone, the car or a bike computer.

Go on a hunt for big numbers: house prices in a folder, the number of visitors at a football match, the kilometre counter (*kilometerstand*) in the car, the number of steps on a step counter, population signs at the edge of a town.

Write five on paper. Read them out loud together. Put them in order. Which is the biggest? Which is closest to 10 000?

Talk about it: "Which of these numbers do you think is rounded? How can you tell?" (Numbers ending in lots of zeros are often rounded.)

2. The money bank

You need: play money or paper slips labelled €10 000, €1 000, €100, €10 and €1 (about ten of each).

1. You say a number, e.g. €23 415. He builds it with the slips: 2 × €10 000, 3 × €1 000, 4 × €100, 1 × €10, 5 × €1.
2. Swap: he builds a number, you read it out loud.
3. Exchanging: "I'll pay you €1 000. Can you give me change in €100 slips?" How many? This prepares for *inwisselen* in unit 6.3.
4. "What is the biggest amount you can make with exactly 6 slips? The smallest?"

Talk about it: "Why do we need a zero in €20 405?"

3. Hallway number line

You need: masking tape, sticky notes, a pen.

Stick a long strip of tape on the floor or wall. Label one end 0 and the other end 10 000 (later: 50 000 and 60 000, then 0 and 100 000).

- You write a number on a sticky note; he places it on the line. Then ask: "How did you decide?"
- He says where the halfway point is (5 000). Then quarter points.
- "Stick 7 500 on. Now 7 450. Now 7 550." Which is closer to 8 000?

Measure with a tape measure if you want to check: on a 10-metre hallway line, every metre is 1 000.

Talk about it: "If this line went from 0 to 100 000, where would 10 000 be?"

4. Rounding the shopping

You need: a supermarket receipt or folder.

1. Round each price to the nearest euro. Estimate the total.
2. Then look at bigger things: a bike (€489), a TV (€649), a holiday (€1 850). Round to the nearest €100 and to the nearest €1 000.
3. "When is rounding good enough? When do you need the exact amount?" (Estimating: is there enough money? Exact: paying.)

Talk about it: “€1 850: nearest hundred is €1 900, nearest thousand is €2 000. Which is more useful when deciding if we can afford it?”

5. The kilometre counter

You need: a car’s kilometre counter (*kilometerstand*), a bike computer, or a step counter on a phone.

Write down the reading before a trip and after. How far did you go? What will the counter show after another 100 km? 1 000 km? What happens when it reaches 99 999?

Talk about it: “The counter shows 45 980. After 30 more kilometres, what will it show? Which digits change?” (46 010: three digits change at once.)

Conversation prompts (any time)

- “What’s the biggest number you can think of that has a 7 in the thousands place?”
- “Is 100 000 a lot? 100 000 seconds is about how many days?” (About one day and four hours. Work it out together with a calculator if he’s curious.)
- “Which is more: 10 000 or a thousand hundreds?” (A thousand hundreds is 100 000, so that’s ten times as much.)
- “Where do you see big numbers in real life?”